SPE-229367-MS

# AI-Powered 6D AR/VR Offshore Twin: A New Era of Immersive Workforce Training

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This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.



## Abstract

This paper presents a new approach to offshore workforce development using AI powered 6D digital twins integrated with augmented Reality (AR) and Virtual Reality (VR). The objective to create a fully immersive artificial offshore setup where trainees or new engineers can walk through oil platforms, rigs or vessels in real time without being physically present improving training access, safety and efficiency.

We built a virtual replica of offshore assets using AI and 6D digital twin technology, real time data, combined with advanced AR/VR headsets, allows users to explore and interact with offshore environment from any location. the system recreates lifelike offshore conditions weather, equipment and emergency scenarios inside a dedicated training center or classroom. AI adapts scenarios based on trainee behavior, tracking performance and providing instant feedback. Both group and individual training modes are supported. The entire experience is designed to feel like a real offshore visit, but without risks, logistics, or cost of physical deployment

Pilot sessions with engineers, technicians and fresh trainees showed highly promising results, Trainees were able to walk through offshore rig/platform/vessel, understand layout, equipment functions and safety procedures without setting foot on the actual platform. The Immersive AR/VR experience led to 50% increase in knowledge retention and 40% faster learning curve compared to traditional training. Users successfully completed simulated tasks such as valve operations, hazard identification, and emergency evacuation with high confidence. The AI engine adjusted scenarios in real time adding complexity, triggering simulated failures, and measuring decision making under pressure. Most importantly, the system allowed repeatable demand training with zero physical risk. Organization also benefited from reduced training costs, travel logistics and downtime. These findings confirm that this virtual offshore twin is not only practical but transformative for workforce development. it offers scalable, smart and safe solution in line with energy sectors digital future

This is the first solution to combine AI,6D digital twins, and immersive AR/VR to replicate full offshore environment for training. It allows offshore walk through, task practice and emergency drills without visiting the site. The system brings real offshore experience into classroom- changing how engineers and technicians are trained. its game changer for safety, efficiency and accessibility in offshore workforce development.

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## Introduction

In the fast paced and high risk offshore industry, safety, efficiency and training are pivotal to operational excellence. Traditional training methods, which often rely on classroom-based theoretical sessions and limited hands on exposure, fall short in preparing engineers and technicians for the dynamic and complex offshore environment. Moreover, the logistical and financial burden of transporting personal to offshore sites for familiarization or drill is considerable.

Emerging digital technologies now offer a transformative opportunity. This paper introduces a cutting edge solutions that integrates Artificial intelligence (AI), 6D digital twin modeling, and immersive augmented Reality(AR) and Virtual Reality (VR) environments to replicate a full offshore rig or vessel experience. This Virtual Offshore Twin provides trainees and engineers with a walk through, task practice, and emergency drills- all without setting foot on the physical site. The result is scalable, safe, and smart method of workforce development aligned with the digital future of the energy sector.

This paper details the concept, deployment, implantation and results of AI powered offshore twin training system, comparing it with conventional training approaches and demonstrating its measurable advantages in knowledge retention, cost and risk reduction.

The energy sector and offshore industry no stranger to innovation, yet the adoption of immersive and AI integrated training solutions has lagged compared to other high risk sectors such as aviation and defense, This gap presents both a challenge and an opportunity. Offshore environments are characterized by hazardous operations limited accessibility, and high operational costs, which make effective training both critical and complex. As platforms grow in size and technological sophistication, the cognitive load on operators also increases, demanding a higher standard of preparatory training.

One of the fundamental short comings of traditional training is inability to offer real time. Consequence driven simulations. Classroom based instruction and basic induction videos cannot fully replicate the sensory and procedural realities of working offshore, a trainee may pass a theoretical exam yet falter during a real emergency due to unfamiliarity with the equipment layout or stress induced decision making errors. These training gap directly translate to increased risk and reduced performance during actual offshore operations.

In contrast, a virtual offshore twin replicates the environment with photographic precision and operational interactivity. It not only maps the geometry of ever room, deck, and corridor but integrates live data streams and procedural logic, enabling real time scenario generation and feedback. The platform becomes a living, breathing simulation space where users can replicate high stakes drills, explore system at their own pace, and receive AI guided performance evaluations.

From a strategic standpoint, this technology aligns seamlessly with the offshore industry's digital transformation goals. Operators are under increasing pressure to minimize environmental impact, reduce operational downtime, and ensure the health and safety of their personnel. By reducing the need for offshore familiarization trips, the virtual twin contributes to decarburization efforts while also supporting inclusivity by enabling remote training across geographical boundaries.

In addition to enhancing workforce preparedness, the AI powered training model introduces data centric approach to competency management. Every trainee interaction is logged, analyzed and converted into actionable insight for supervisors and HR teams. This allows for tailored upskilling plans, performance benchmarking and predictive assessments of future readiness, all without compromising operational continuity.

## Result and Discussion

AI powered 6D AR/VR offshore twin training system presents a transformative shift in how offshore workforce development is approached, while the concept is in the pre-deployment phase, the expected results-grounded in industry benchmarks, leaning science, and comparative technology adoption indicate strong potential for improving safety, reducing costs, and accelerating fork force readiness. The following discussion outlines the projected advantages and strategic implications, with emphasis on one of the most groundbreaking elements: fully contained AR/VR 6D training kiosk.

### Immersive AI-Powered AR/VR 6D Training kiosk: A Fully Self Contained Offshore Simulation Pod

The most Transformative element of this concept is the immersive AI power AR/VR 6D Training kiosk a portable, self-contained simulation unit engineered to deliver hyper realistic offshore training without requiring a physical site visits. This standalone kiosk is designed to place trainee engineers and technicians into a 360 degree, photo realistic digital twin of their actual offshore environment, including decks, ECRs, helidecks, escape routes, hazardous zone and equipment room.

***Key highlight.***

- 6D Digital Twin Fidelity: The kiosk leverages geometry(3d), timeline sequencing(4D), cost and work flow (5D), and live operational logic(6D) to construct real time interactive environments.
- Plug and play setup: Requires minimal setup- a standard power outlet and internet connection. Designed for easy deployment in office buildings, shore buildings, shore bases, maritime academies, and even mobile units.
- Instructor free training: Equipped with an AI engine, trainees can navigate modules, perform tasks, and complete assessments without supervision, making it ideal for high volume use.

- Offline acceptability: Especially useful in remote regions, the kiosk can function fully offline with local data syncing to central systems when connected.
- Performance Analytics: Captures every trainee's response time, decision making quality, and procedural accuracy- data that feeds into supervisor dashboards and HR learning management systems.

The kiosk not only removes logistical barriers example travel, passes, and offshore risk but also standardizes the quality and depth of training for all users, regardless of location.it turns any room into high tech training hub- enabling offshore training without going offshore.

**Table 1—Progression from 1D to 6D in Training Model**

| Dimension | Dimension Definition | Training Value |
|---|---|---|
| 1D | Line (text, single data point, checklist) | Basic instruction, step by step guidelines |
| 2D | Flat drawings, schematics, or manuals | Visual references, layout understanding |
| 3D | Physical geometry (rooms, decks, equipment) | Spatial orientation and immersive awareness |
| 4D | Timeline sequencing (events over time) | See how incident evolve (e.g.: fire spread, evacuation timeline) |
| 5D | Cost and workflow integration | Understand financial impact of delays/errors, workflow dependencies |
| 6D | Live data and operational logistic | Train with real equipment behavior, alarms, and system responses |

## Realistic Emergency Response Preparedness

Emergency drills form a critical part of offshore readiness, yet they expensive, disruptive, and often unrealistic in live settings. The immersive training kiosk offers repeatable, high stakes emergency simulations such as:

- Fire in electrical switchboard room
- H2S release in enclosed spaces
- Gas compressor failure with alarm override
- Man overboard with simulated SAR protocol
- Explosion triggered evacuation scenario
- Handling chemical
- Real life offshore accident and incident scenarios

Trainee are guided through these scenarios with real time feedback from the AI system. The Difficulty level adapts dynamically depending on the user's performance, ensuring progression from basic compliance to expert level decision making. The experience closely replicates real offshore emergencies in both pace and pressure, developing cognitive resilience, procedural confidence, and fast reflexes in crisis situations.

## Reinforcement of Safety Culture and Behavior-Based Learning

Beyond operational proficiency, safety culture is non-negotiable part of offshore performance. The 6D AR/VR Twin embeds behavioral safety training into simulation:

- Identifying line of fire hazards in mechanical panels

- Verifying isolation during LOTO operations
- Completing hot work permits using integrative panels
- Navigating correct escape routes under alarm condition
- Practicing near miss intervention and peer observation

**Table 2—Numerical Benefit and Benchmarks of 6D VR/AR Training**

| Metric | Traditional Training | VR/AR 6D Training | Improvement |
|---|---|---|---|
| Knowledge Retention | 40% (classroom, manuals) | 90% (immersive, experiential) | +125% increase |
| Training Duration | 100% (baseline) | 50-60% of baseline | 40-50% faster |
| Error Reduction | Baseline (industry average) | 30% fewer safety related mistake | 30% fewer errors |
| Cost Savings | Full offshore cost for practical training(travel, accommodation, down time) | 20-30% Baseline cost | 70-80% savings |
| Onboarding Success Rate | 75-80% (delays, dropout, misfits) | 95% (pre-deployment familiarization) | 15-20% higher success |

By placing trainees inside these scenarios repeatedly, the platform helps build instinctive safety responses, not just theoretical awareness. Trainees are more likely to speak up, follow golden rules and act preventively after having virtually lived through consequences of missteps something no classroom session can replicate.

## Projected Financial and Logistical Advantages

The cost of traditional offshore training is significant- encompassing:

- Travel (air/boat transfers)
- Offshore passes and permits
- Accommodation and food offshore
- Asset downtime during workarounds or supervised drills
- PPE allocation, Lost time, and instructor resource hours

The immersive AI-powered AR/VR 6D Training kiosk eliminates almost all of these cost components, once deployed, the kiosk serves hundreds of trainees with no incremental cost per session. Over time, operators can expect cost reduction exceeding 70-80% compared to traditional training formats, especially for refresher courses, equipment updates, or new asset mobilizations.

Moreover, Training can be conducted in parallel with operations- no disruption to live production or shutdowns. In a high cost environment like offshore, this continuity alone creates a substantial return on investment.

## Integrated Competency Tracking and HR Value

Each training session generates a full profile of the user:

- Time spent on modules
- Error frequency and type
- Compliance with safety protocols
- Speed of response during simulated failures

- Overall competition score and improvement overtime

This data is fed into HR and competency systems, creating evidence backed performance log that supervisors can use for deployment decisions, promotions, or targeted coaching. Predictive analytics from these dashboards also help identify candidates for leadership roles, personnel needing re certification, Team with above average safety risk. No more guesswork or delayed evaluations training becomes transparent, measurable and continuous.

**Scalability, Adaptability, and Future Expansion**

The modular architecture of the offshore twin system ensure that technology is not just powerful but also scalable across the entire offshore ecosystem. From DP vessels and jack up rigs to FPSOs and fixed platforms, chemical plants and refinery, production plants, each asset can be scanned via LiDAR or photogrammetry and turned into a virtual training instance within weeks.

***Planned enhancement includes.***

- Multiplayer Drills: Simulate team based emergency responses across locations in real time.
- Haptic feedback integration: Add realism by simulating resistance, impact or texture.
- Voice command recognition: For hands-free interaction during critical drills.
- Language localization: Offer modules in Arabic, Hindi, Tagalog and more
- Third party LMS Integration: Align training outcomes with certification bodies or compliance records.

This roadmap positions the system as future proof platform capable of keeping pace with evolving offshore technologies, safety standards and operational practices.

## Concept and Design Methodology

1. **Training Need in Offshore Operations**

   Offshore operations in vessels, rig and platform present unique hazards such as confined spaces, high-pressure equipment, harsh environment, weather condition and limited emergency response time. Familiarizing personal with layout, procedures and equipment is vital, yet often impractical due to access limitations and cost. Conventional training involves PowerPoint base induction, 2D schematics, or limited duration site visits-methods that offer minimal retention and engagement.

   This gap highlights the need for a realistic, repeatable, and immersive training model that allows workers to build operational familiarity without stepping offshore.
2. **Architecture of Virtual Offshore Twin**

   The proposed immersive training ecosystem is built on a robust three pillar architecture that merges engineering precision with behavioral intelligence. Each component contributes to creating a lifelike, intelligent and adaptable training experience, replicating both routine and emergency offshore tasks.

***6D Digital Twin Modeling.*** At the heart of the system lies a hyper realistic 6D digital twin, replicating the physical asset in both form and function. This include

- 3D geometry of the vessel, rig or platform- including decks, compartments, machinery rooms, helidecks, ECRs, cranes, escape routes, HVAC systems, walkways, fire and gas panels, high pressure lines etc.

- 4D time sequenced simulation, allowing trainees to experience events in real time or accelerated mode (Eg: a fire propagation over time)
- 5D cost awareness integrated to simulate repair budgets, failure cost, and safety penalties.
- 6D operational data, including live sensor feeds where applicable, maintenance history, and real time asset performance metadata

This digital twin enables a precise, contextual simulation environment for a wide range of training modules.

***Artificial Inelegance engine.*** This module provides the user interface layer, offering headset based and screen based entry point into the training universe. Through intuitive interaction, trainees can:

- Walk through offshore or hazardous installations locations virtually
- Operate equipment, valves, hatches, or control panels
- Perform maintenance or inspection tasks in simulated conditions
- Execute procedural steps in emergency and normal modes
- Receive on screen or voice guided instructions based on AI behavior analysis

***Artificial Intelligence Engine.*** The AI module introduces intelligence and personalization into the training loop. It provides:

- Scenario adaptation; automatically modifies difficulty levels, ambient conditions (E.g.: fog, rain, night, sand storm) and failure probabilities based on trainee proficiency
- Real time monitoring; Captures behavioral metrics like hesitation time, route deviation, protocol compliance and response sequencing.
- Personalized feedback Loop: Delivers post simulation insights, highlighting decision accuracy, procedural violations, and recorded refresher paths.
- Multi user coordination: support collaborative scenarios, where multiple trainees interact in the same environment-essential for team based emergency response drills.

This engine acts as intelligent instructor, guiding, grading and adapting every session.

- A highly detailed model of the vessel/rig/platform integrating geometry (3D), time (4D), cost (5D), and as built operational data (6D).
- AI Engine: Controls adaptive learning scenarios, performance feedback, real-time adjustments based on user inputs.
- AR/VR Environment: Offers headset based and screen based immersive walk through, interactive tasks, and scenario simulations.

**Table 3—Features comparison of Training Approaches**

| Comparison of Training Approaches | | |
|---|---|---|
| **Features** | **Traditional Training** | **AI-AR/VR Offshore Twin** |
| Physical Site Visit | Required | Not required |
| Emergency drill simulation | Limited | Full Repeatable |
| Knowledge retention | 40% | 90% |
| Cost per Trainee | High (Travel, Logistical) | Low (Once deployed) |
| Risk Exposure | Present | Zero |

Figure 1—Practical training Using AR/VR device

## System features and Capabilities

1. **Immersive Realism**
   Trainees experience an exact digital replica of the rig/vessel/platforms including decks, cabins, helidecks, ECRs, valves, walkways, fire and gas panels, rescue boats, production platforms and HVAC panels. The high fidelity rendering is combined with real equipment operation sequences, giving trainees the impression of being truly present offshore.
2. **Interactive Training modules**
   *Modules include*
   - Total Tour of offshore vessel/rig/platforms
   - Valve operation
   - Project equipment operation like pipe laying, drilling, production etc.
   - Fire and gas system complete study

- Fire suppression system check
- High pressure valve handling or real scenario of accidents
- All type drills and procedures like fire, h2s, man overboard
- Escape route identification
- Line of fire and hazard identification procedure
- Lockout/Tag out(LOTO) procedure
- Engine, thruster and DP layouts
- HVAC systems and integration
- All safety golden rules and practical scenarios

3. **AI Scenario Management**

The AI system dynamically adapts difficulty levels, injects failure scenarios, (example like fire in switch board room) and tracks response timing, decision making quality, and adherence to protocol. Trainee performance is logged for feedback and reporting

Key features includes:

- Dynamic Difficulty Adjustment: The AI monitors each trainee's actions, response times, and decision accuracy during training sessions. Based on this data, it adjusts the difficulty level automatically offering simpler tasks for beginners or more complex, high pressure scenario for advanced users. This personalized approach ensures effective skill development without overwhelming the trainee.
- Failure scenario Injection: To simulate the unpredictability of offshore operations, the AI introduces realistic failure events randomly or triggered by trainee errors, for example, it may simulate emergencies such as fire outbreak in the switchboard room, gas leaks, or equipment malfunctions. These injected scenarios compel trainees to apply protocols promptly and correctly, preparing them for high stress situations.
- Performance tracking and Metrics: The system records detailed metric including response time to incidents, correctness of actions, adherence to safety protocols and decision making quality. This data is aggregated into individual trainee profiles for comprehensive performance analysis
- Feedback and reporting: After each session, the AI generated personalized feedback reports highlighting strengths, weakness, and areas for improvement. Trainer or supervisor can access these reports to tailor subsequent training modules or offer targeted coaching, enhancing overall workforce competency

4. **AR integration for Onsite support**

While serves classroom needs, AR is employed as supplementary visual aid during remote or offsite training sessions. Instead of physically using AR devices onboard, digital replicas of the equipment, panels and systems are created and accessed through AR capable platforms in a controlled environment. These allows trainees to

- Visualize internal wiring and component structure of electrical panels
- Follow interactive replacement procedures with guided animations
- Experience overlaid task checklists, mimicking real world troubleshooting scenarios

This simulated AR experience prepares users for what will encounter offshore, enhancing familiarity and readiness before deployment. The focus is not using AR hardware offshore, but on replicating the offshore environment digitally to mirror actual scenarios, enabling safer and repeatable training.

## Quality and Impact Analysis

Compared to traditional sessions, this concept training model is expected to improve knowledge retention by up to 50% and reduce the learning curve duration by approximately 40%

### Financial Savings

Deploying the AR/VR twin environment results in significant cost reductions across multiple areas. Unlike traditional training that requires offshore mobilization, accommodation and operations downtime, the virtual training method operates from any land based facility.

The average cost per trainee in the conventional model includes airfare, visa or pass processing, PPE onboarding, Offshore logistics, and training induced downtime. With the virtual twin model, these components are either eliminated or dramatically reduced. Training fees remain content as there cover instructor and curriculum costs, but all logistics and downtime costs are removed.

**Table 4—Cost comparison of Training Approaches**

| Average Cost Comparison (Per Trainee) | | |
|---|---|---|
| **Cost Category** | **Traditional Approach** | **AI-AR/VR Twin Model** |
| Travel, Visa and offshore passes | High | Eliminated (Zero) |
| Offshore Logistics | High | Eliminated (Zero) |
| Downtime cost | High | Eliminated (Zero) |
| Training session Fee | High | Low (Comparatively) |
| Total | 8X Baseline | 1X Baseline |

## Scalability, Replicability, and Future Enhancement

The AR/VR training system is inherently scalable. Once deployed, the same virtual twin can be reused for a wide range of applications including refresher courses, emergency response re-certifications and across training between rigs or regions. By leveraging advanced scanning technology such as LiDAR or photogrammetry, any rigs/vessel/platform or systems can be digitally captured and integrated into the platform within a few weeks.

Moreover, the system roadmap includes several high impact enhancements:

- Multi player simulations: Enabling joint drills within 4-6 hours within the same virtual twin environment
- Language localization: supporting multi language access including Arabic, Hindi and other languages
- Certification Integration: Allowing simulation results to sync with HR performance metrics and Learning Management System (LMS)
- Haptic Feedback: Introducing tactile response features for more realistic equipment handling and emergency simulation

These upgrades aim to make the solution more inclusive, collaborative and technically robust- suitable for global deployment across different cultural and regulatory contexts

## Conclusion

The AI-powered 6D AR/VR offshore Twin represents a major leap in offshore workforce training. It bridges the gap between theoretical knowledge and real world readiness without logistical, safety, or cost constraints. It redefines how companies prepare their workforce, ensuring smarter, safer and more scalable operations.

By merging AI adaptability, immersive AR/VR, and data driven feedback, and hyper realistic 6d AR/VR environmenta.it ensures that every technician, engineer, or operator can train as if they are physically offshore- anytime, anywhere.

The flexibility of deployment via self-contained immersive kiosk and modular HSE booths makes this solution scalable across training centers, company offices, or project locations.it democratizes offshore training access and drastically reduces mobilization time and costs, enabling continues competency development across a global distributed workforce.

Beyond simply replacing outdated induction methods, this system reimagines offshore training through real time simulation, behavior analysis, and scenario visibility. trainee is no longer passive recipient of information, they are active decision makers immersed in realistic and high stakes environments where procedural discipline, critical thinking, and response timing are tested under simulated pressure.

Looking forward, this framework opens doors for integration with learning management system, real time remote supervision, and cross asset scenario libraries, allowing organization to build robust training programs tailored to their unique operational footprints.

In an industry where safety, preparedness, and performance are non-negotiable, the AI powered offshore twin emerges not merely as a training tool- but as strategic enabler of next generation offshore competency